Ben Potter

bp@benpot.com — Queen's University

PERSONAL SUMMARY

Ben Potter is a Master's student in Computer Engineering at Queen's University researching bio-inspired polarization compass systems. His work on insect-inspired navigation has led to multiple publications, including a journal paper in IEEE Access. Ben combines expertise in computer vision, embedded systems, and hardware development to advance autonomous navigation technologies. He previously completed his undergraduate degree with first class honours and served as Chair of the IEEE Kingston Section Student Branch.

TECHNICAL SKILLS

Programming Languages: Rust, Python, Java, JavaScript, C/C++, MATLAB, LaTeX, R
Hardware & Embedded: Raspberry Pi, Arduino, Industrial Cameras, Linux
Software Engineering: Git, Jira, Agile Methodologies, Software Architecture, OpenCV, Robot Operating System (ROS)
Research Tools: Technical Writing, Data Analysis, Academic Publishing, Literature Review

EDUCATION

Queen's University, Kingston, Canada — May 2025 — Present
Master of Applied Science in Computer Engineering — GPA: 4.00
Supervisor: Dr. Muhammad Alam
Supervisor: Dr. Yahia Antar
Interests: Polarization-based Navigation; Fused-sensor Systems; Safety-critical Software

Queen's University, Kingston, Canada — Sep. 2021 — Apr. 2025
Bachelor of Applied Science in Computer Engineering — First Class Honours, GPA: 3.71

PUBLICATIONS

[1] D. Agarwal, B. Potter, J. Y. Siddiqui, Y. Antar, and M. Z. Alam. Bio-inspired polarization compass for solar azimuth prediction. In *2024 Photonics North (PN)*, pages 1–2, 2024.

[2] D. Agarwal, B. Potter, J. Yaseen Siddiqui, Y. M. M. Antar, and M. Z. Alam. Bio-inspired polarization compass for solar azimuth prediction under clear and cloudy sky conditions. *IEEE Access*, 13:3242–3250, 2025.

[3] B. Potter, D. Agarwal, Y. Antar, and M. Z. Alam. Evolution of hough transform for solar azimuth prediction. In *2024 IEEE Canadian Conference on Electrical and Computer Engineering (CCECE)*, pages 186–187, 2024.

[4] B. Potter, D. Agarwal, J. Y. Siddiqui, Y. Antar, and M. Z. Alam. Bio-inspired polarization compass. In *2025 Inquiry at Queen's*, 2025.

ACADEMIC EXPERIENCE

Automated Planning for Air Traffic Control
Master's Coursework — Sep. 2025 — Dec. 2025
Instructor: Dr. Jon Gammell

- Identified the systematic manpower shortage in current air traffic control infrastructure.
- Implemented and extended novel research on automated air traffic control planning systems.
- Developed an academic report that outlined my findings.

Radiation-Induced Anomaly Detection in System Call Logs
Bachelor's Thesis — Sep. 2024 — Dec. 2025
Supervisor: Dr. Sean Kauffman

- Analysed over 10 billion system call logs generated by embedded devices under radiation beam.
- Discovered an anomaly that occurred before complete system failure.
- Developed an academic report that discussed our findings.

Bio-Inspired Polarization Compass
Junior Research Assistant, Royal Military College of Canada — Sep. 2023 — Apr. 2025

- Developed camera-based hardware for capturing skylight polarization images.

- Implemented novel solar azimuth parsing algorithms.
- Three publications including one journal paper.
- Presented work in multiple academic venues.

WORK EXPERIENCE

QA Consultants
Associate Intern Mar. 2022 — Sep. 2022

- Worked in an agile team of five focused on R&D for automated software testing.
- Delivered brand new model-based testing software to a large client with close to 4,000 employees.
- Retrofitted existing testing architecture that drastically improved maintainability and reduced total line count by 50%.
- Worked with Java, Python, Selenium, Git, Jira.

Hatch Coding
Teacher Sep. 2019 — Sep. 2021

- *Course delivery:* Taught Javascript and Python to students 10-15 years old.
- *Couse development:* Developed course material for intermediate and advanced programming classes.

VOLUNTEER EXPERIENCE

Institute of Electrical and Electronics Engineering
Chair, Kingston Section Student Branch Jan. 2024 — Dec. 2026

- Tripled the number of active members during my two year tenure.
- Organized three student competitions with cumulative prize pools of $1,500.
- Organized technical lectures from distinguished speakers at Google, NASA, and Queen's.

Queen's University
SGPS Representative, Graduate ECE Council May 2025 — Apr. 2026

AWARDS

- Vice-Principal Undergraduate Research Award, $1000 CAD
- NSERC Undergraduate Student Research Award, $7000 CAD
- Favourite Graduate TA, Academic Year 2025-26